HW 3

Due: Tue Sep 30, 2025 11:59pm

1. Suppose that $x > 0, y < 0$. Using definition of absolute value and splitting into cases, show that $|x + y| \leq |x - y|$.

2. Suppose you have an iterative algorithm for finding a fixed point of a function F(x) on a certain interval. Write a short text (2-4 paragraphs) explaining the following.

   1) How do you decide when to stop iterating?

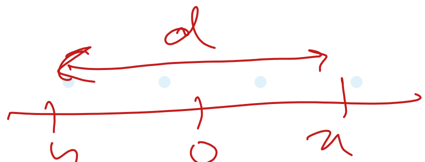
   2) What is your strategy for making sure that the computed solution is close to the exact solution?

DO NOT use the standard methodology and estimates from the textbook. Try to find other ways of accomplishing the goals above. Discuss briefly the advantages and disadvantages of your error control approach.

3. Do Problem 23 c) from Section 3.4.

4. Do Problem 32 from Section 3.4.

1. Suppose that $x > 0, y < 0$. Using definition of absolute value and splitting into cases, show that $|x+y| \le |x-y|$.



Given $x>0, y<0$

Assume $x-y=d,\ d>0$

$\therefore$ RHS $= |x-y| = d$ is always true.

$x+y = 2y+d$

(I) $x+y \geq 0 \Rightarrow 2y+d \geq 0$

LHS: $2y+d$ RHS: d

As $2y<0 \Rightarrow$ LHS $<$ RHS

(II) $x+y<0 \Rightarrow x+(x-d)<0 \Rightarrow 2x-d<0$

5 −8

$d=13$ $\Rightarrow |x+y| = |x+(x-d)|$

LHS $= -2x+d$ RHS $= d$

$\because -2x<0 \Rightarrow$ LHS $<$ RHS

$\because$ Cases (I) and (II) exhaustively cover all cases and LHS $<$ RHS in both

$\therefore$ $\boxed{|x+y| < |x-y| \ \forall\ x>0, y<0}$ $\square$

2. Suppose you have an iterative algorithm for finding a fixed point of a function F(x) on a certain interval. Write a short text (2-4 paragraphs) explaining the following.

1) How do you decide when to stop iterating?

2) What is your strategy for making sure that the computed solution is close to the exact solution?

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Assumed / Given:

1. $F(x)$ is a contractive function in the given interval, and we know $F(x)$.
2. $F(x)$ has a fixed point α (unknown) in the given interval.

Approach:

Step 1 From knowledge of $F(x)$, I can determine its order of convergence via Taylor's formula. Let's say I get the order of convergence as p. (1)

Step 2 I can generate a table of n vs x_n values by recursively using $x_{n+1} = F(x_n)$.

n	x_n
0	x_0
1	x_1
2	x_2
⋮	⋮
n	x_n
⋮	⋮

∵ order of convergence = p.

As x_n approaches r,

$$\left|\frac{\varepsilon_{n+1}}{\varepsilon_n^{\,p}}\right| \to C$$

or
$$\left|\frac{x_{n+1}-r}{(x_n-r)^p}\right| \to C$$

$$(x_{n+1}-r) = C(x_n-r)^p$$

$$\log(x_{n+1}-r) = \log C + p\log(x_n-r)$$

$$\log(x_n-r) = \log C + p\log(x_{n-1}-r)$$

3. Do Problem 23 c) from Section 3.4.

PROBLEMS 3.4

23. Find the order of convergence of these sequences.

c. $x_n = (1 + 1/n)^{\frac{1}{2}}$

$r = 1$

$e_n = x_n - r = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}} - 1$

$e_{n+1} = x_{n+1} - r = \left(1 + \frac{1}{n+1}\right)^{\frac{1}{2}} - 1$

$(e_{n+1})^2 = \left(1 + \frac{1}{n}\right)$

$(e_n + 1)^2 - 1 = \frac{1}{n}$

$e_n(e_n + 2) = \frac{1}{n}$

$e_{n+1} = \left(1 + \frac{1/n}{1 + 1/n}\right)^{\frac{1}{2}} - 1$

or $$\rho_{n+1} = \left(1 + \frac{\rho_n(\rho_n+2)}{\rho_n(\rho_n+2)+1}\right)^{\frac{1}{2}} - 1$$

or $$(\rho_{n+1}+1)^2 - 1 = \frac{\rho_n(\rho_n+2)}{\rho_n(\rho_n+2)+1}$$

or $$(\rho_{n+1})(\rho_{n+1}+2) = \frac{\rho_n(\rho_n+2)}{(\rho_n+1)^2}$$

Try $\rho_{n+1} = c\,\rho_n$:

$x = \rho_n$

$$(cx)(cx+2) = \frac{x(x+2)}{(x+1)^2}$$

$$\cancel{c^2(x)(x+2)} = \frac{\cancel{x(x+2)}}{(x+1)^2}$$

$$c^2(x+1)^2 = 1$$

$$c = \frac{1}{x+1}$$

Can't be dependent on x!

Try $e_{n+1} = C e_n^p$:

$(C k^p)(C k^p + 2)(C k+1)^2 \ldots k(k+2)$

$2p+2 \ldots 2$

$p = 0$?

$$x_n = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

$$r = \lim_{n \to \infty} x_n = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^{\frac{1}{2}} = 1$$

$$e_n = x_n - r = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}} - 1$$

$$e_{n+1} = x_{n+1} - r = \left(1 + \frac{1}{n+1}\right)^{\frac{1}{2}} - 1$$

$$(e_n + 1) = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

$$\text{or } (e_n + 1)^2 = 1 + \frac{1}{n}$$

$$\text{or } (e_n + 1)^2 - 1 = \frac{1}{n}$$

$$\text{or } \frac{1}{(e_n + 1)^2 - 1} = n$$

$$\Rightarrow e_{(n+1)} = \left(1 + \frac{1}{\frac{1}{(e_n + 1)^2 - 1} + 1}\right)^{\frac{1}{2}} - 1$$

$$\Leftrightarrow \quad \rho_{n+1} = \left(1 + \frac{(\rho_n+1)^2 - 1}{1 + \left((\rho_n+1)^2 - 1 \right)} \right)^{\frac{1}{2}} - 1$$

$$\Leftrightarrow \quad \rho_{n+1} = \left(1 + \frac{(\rho_n+1)^2 - 1}{(\rho_n+1)^2} \right)^{\frac{1}{2}} - 1$$

$$\Leftrightarrow \quad \rho_{n+1} = \left(\frac{2(\rho_n+1)^2 - 1}{(\rho_n+1)^2} \right)^{\frac{1}{2}} - 1$$

$$\Leftrightarrow \quad (\rho_{n+1}+1)^2 = \frac{2(\rho_n+1)^2 - 1}{(\rho_n+1)^2}$$

4. Do Problem 32 from Section 3.4.

PROBLEMS 3.4

32. Let $\frac{1}{2} \le q \le 1$, and define $F(x) = 2x - qx^2$. On what interval can it be guaranteed that the method of iteration using F will converge to a fixed point? (This problem is related to Problem **3.2.5**, p. 90.)

PROBLEMS 3.2

5. What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2 y$$

Identify it as the Newton iteration for a certain function.

$$x_{n+1} = 2x_n - yx_n^2 = x_n - (-x_n + yx_n^2)$$

$$\equiv \quad x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$

$$\Rightarrow \quad \frac{f(x)}{f'(x)} = yx_n^2 - x_n$$

$$\frac{f}{\frac{df}{dx}} = yx_n^2 - x_n$$

$$\frac{f}{df} = \frac{yx_n^2 - x_n}{dx}$$

$$\frac{df}{f} = \frac{dx}{yx_n^2 - x_n}$$

$$\Rightarrow \ln f(x_n) = \int \frac{dx_n}{y x_n^2 - x_n}$$

$$= \int \frac{1}{x_n(y x_n - 1)}\, dx_n$$

$$\frac{1}{y} \int \frac{1}{x_n(x_n - 1/y)}\, dx_n$$

$$\int \left(\frac{1}{x_n - 1/y} - \frac{1}{x_n}\right) dx_n$$

$$\Rightarrow \ln f(x_n) = \ln\left(x_n - \frac{1}{y}\right) - \ln(x_n) + \ln k$$

$$\Rightarrow \ln f(x) = \ln\left(\frac{x_n - \frac{1}{y}}{x_n} k\right)$$

$$\Rightarrow \boxed{f(x_n) = k\left(1 - \frac{1}{x_n y}\right)} \; ?$$

$$\boxed{r = \frac{1}{y}}$$

$$f \Rightarrow f(x_n) = k\left(1 - \frac{1}{y} x_n^{-1}\right)$$

$$f'(x_n) = k\left(\frac{1}{y} x_n^{-2}\right)$$

$$\frac{f}{f'} = \frac{1 - \frac{1}{y} x_n^{-1}}{\frac{1}{y} x_n^{-2}}$$

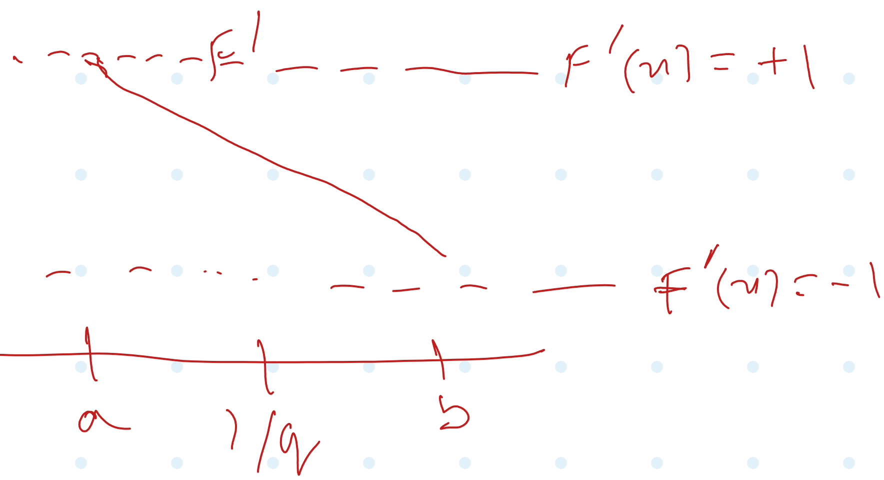
$$\Rightarrow \frac{f}{f'} = \frac{y x_n^2 - x_n}{1}$$

Need to find $[a,b]$, $a < \frac{1}{q} < b$ s.t. $|f'(x)| < 1$ $\forall\, x \in [a,b]$.

$f(x) = 2x - qx^2$

$f'(x) = 2 - 2qx$ → for $q > 0$, this is monotonically decreasing.

$\Rightarrow$



$\Rightarrow$ $f'(a) < 1$ && $f'(b) > -1$

or $2 - 2qa < 1$ && $2 - 2qb > -1$

or $\boxed{a > \frac{1}{2q}}$ && $\boxed{b < \frac{3}{2q}}$

$\Rightarrow$ $F(x)$ will converge to a fix point in $\left(\frac{1}{2q}, \frac{3}{2q}\right)$

This interval widen as $q \downarrow$ $\Rightarrow$ Choose max. value of q to get common denominator when

$$\Rightarrow \boxed{x \in \left(\frac{1}{2}, \frac{3}{2}\right) \text{ for convergence of } F(x) \text{ to fixed point.}}$$